1. **Extract B** (lines 8–10) states: 'many companies intentionally make it difficult to repair their products in order to increase sales. This is both economically inefficient and environmentally foolish; it imposes costs on the environment even though it may make sense for individual companies.'

With the help of a diagram, explain how the production of goods which are designed not to last long may result in market failure.

[9 marks]

2. At current levels of output, the marginal social cost of making a good is greater than its marginal private cost. Also, the marginal social benefit of the good is greater than its marginal private benefit. As a result, there will probably be

A a misallocation of resources.

B positive externalities in production.

C negative externalities in consumption.

D economies of scale.

[1 mark]

3. Explain what is meant by 'property rights' (**The Global Context Extract B**, line 20) **and** analyse how an absence of property rights can worsen the environmental market failures of **both** pollution **and** the rapid depletion of natural resources.

[9 marks]

4. Which combination of example and policy, **A**, **B**, **C** or **D**, is most likely to be consistent with the source of market failure identified?

	Source of market failure	**Example of market failure**	**Policy to correct market failure**
A	Missing market	Health care	State provision
B	Positive consumption externalities	Education	Minimum price
C	Merit good	Petrol	Indirect taxation
D	Negative production externalities	Electricity generation	Pollution permits

[1 mark]

5.

THE GLOBAL CONTEXT

Extract B: Climate change

In 2004, Russia signed up to the Kyoto Protocol, an international agreement on policies to manage climate change. This committed Russia to a United Nations programme to reduce emissions of greenhouse gases. In 2010, Russia's emissions were as low as their 1990 levels, but this was less to do with environmental policy and more to do with the collapse of heavy industry following the fall of the Soviet Union. A large contribution towards the UK's Kyoto targets also came from deindustrialisation. However, China, like the USA, did not sign up to the Kyoto Protocol.

International action to control carbon emissions is necessary. One policy is known as 'carbon trading'. Countries agree overall emissions limits and try to create an international market for permits to pollute. The idea is to put a money value on emissions and, over time, to create incentives for businesses to invest in less-polluting technology. As well as the large compulsory 'compliance' market for carbon credits, there is also a smaller voluntary market for 'carbon offsets'.

For example, an organisation can be paid to plant trees to allegedly offset the damage that an airline journey does to the ozone layer. Some companies are using these offsets, together with greater energy efficiency and reduced polluting activities, to claim to be 'carbon neutral'. Some environmentalists are suspicious of offset schemes and claim, for instance, that offset schemes have hidden environmental costs.

Some economists argue that the pollution and depletion of communal resources such as clean air, water reserves and fish stocks are made worse by a lack of property rights. Critics say that carbon trading amounts to 'privatisation of the atmosphere', with property rights in this vast new market being grabbed by speculators or by those who already pollute the most. They also claim that activities such as buying credits from Russia are an 'accounting fraud' because they do not necessarily arise from energy efficiency but from the collapse of heavy industry. There is concern that carbon trading is a further blow to manufacturing and that service industries are treated more leniently. If the UK relies on manufactured imports, it can be argued that factories located in China are actually creating UK emissions.

Some major UK businesses appear happier with permit trading than with alternative policies such as quotas on emissions or the taxation of fuel. There is a suspicion that carbon trading might mislead the public into believing that pollution from airlines and other industries is no longer damaging. Governments are accused of being overgenerous in allotting permits to industries that they see as nationally important, such as car manufacturing and electricity generation. While the UK might reduce its domestic demand for carbon, it continues to contribute to the world supply of fossil fuels through activities such as oil prospecting and opencast mining. Russia, China and less-developed countries with coal reserves show few signs of reducing their output of this form of carbon.

Source: news reports, 2010

6.

Pollution and Climate Change Extract F (lines 9−10) states 'Some major UK businesses appear happier with permit trading than with alternative policies such as quotas on emissions or the taxation of fuel'.

Using the data in the extracts and your knowledge of economics, evaluate the possible impact on UK businesses of policies to reduce pollution.

[25 marks]

7. Critics of the world's five most valuable multinational technology firms (Google, Amazon, Apple, Facebook and Microsoft) argue that there ought to be greater government intervention to protect consumers' interests. Several European governments are considering imposing new taxes on the revenues of such firms, rather than their profits.

Discuss whether governments should consider increasing the regulation and taxation of technology firms which have acquired significant global monopoly power.

[25 marks]

8.

Pollution and climate change

Extract F: Issues over carbon trading

Critics say that carbon trading amounts to 'privatisation of the atmosphere', with property rights in this vast new market being grabbed by speculators or by those who already pollute the most. They also claim that activities such as buying credits from Russia are an 'accounting fraud' because they do not necessarily arise from energy efficiency but from the collapse of heavy industry. There is concern that carbon trading is a further blow to manufacturing and that service industries are treated more leniently. If the UK relies on manufactured imports, it can be argued that factories located in China are actually creating UK emissions.

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Source: news reports, 2010

9. In December 2018, the Competition and Markets Authority (CMA) announced a range of policies to prevent firms charging existing customers more than new customers. Firms supplying financial services, mobile phones and broadband are no longer allowed to discriminate against loyal customers renewing their contracts.

Assess the view that price discrimination is always damaging.

[25 marks]

10. In December 2018, the Competition and Markets Authority (CMA) announced a range of policies to prevent firms charging existing customers more than new customers. Firms supplying financial services, mobile phones and broadband are no longer allowed to discriminate against loyal customers renewing their contracts.

Explain why imperfect information can lead to market failure.

[15 marks]